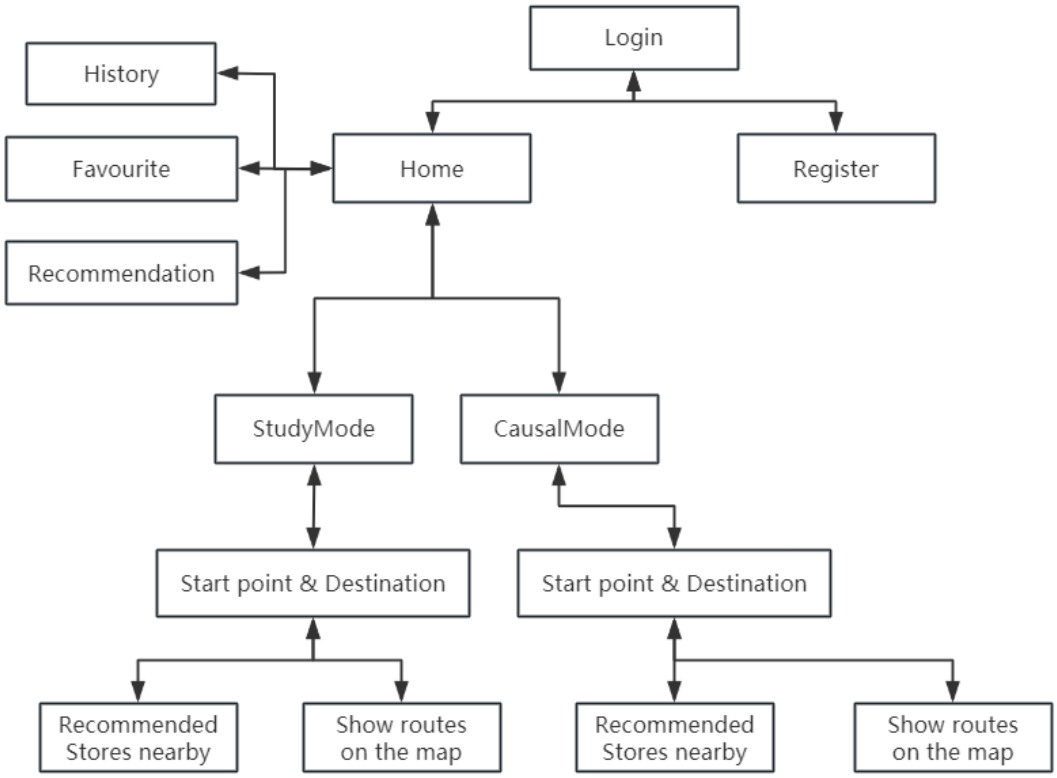
Recommendation System for Maps' Guided System Based on API

Criterion B – Designing

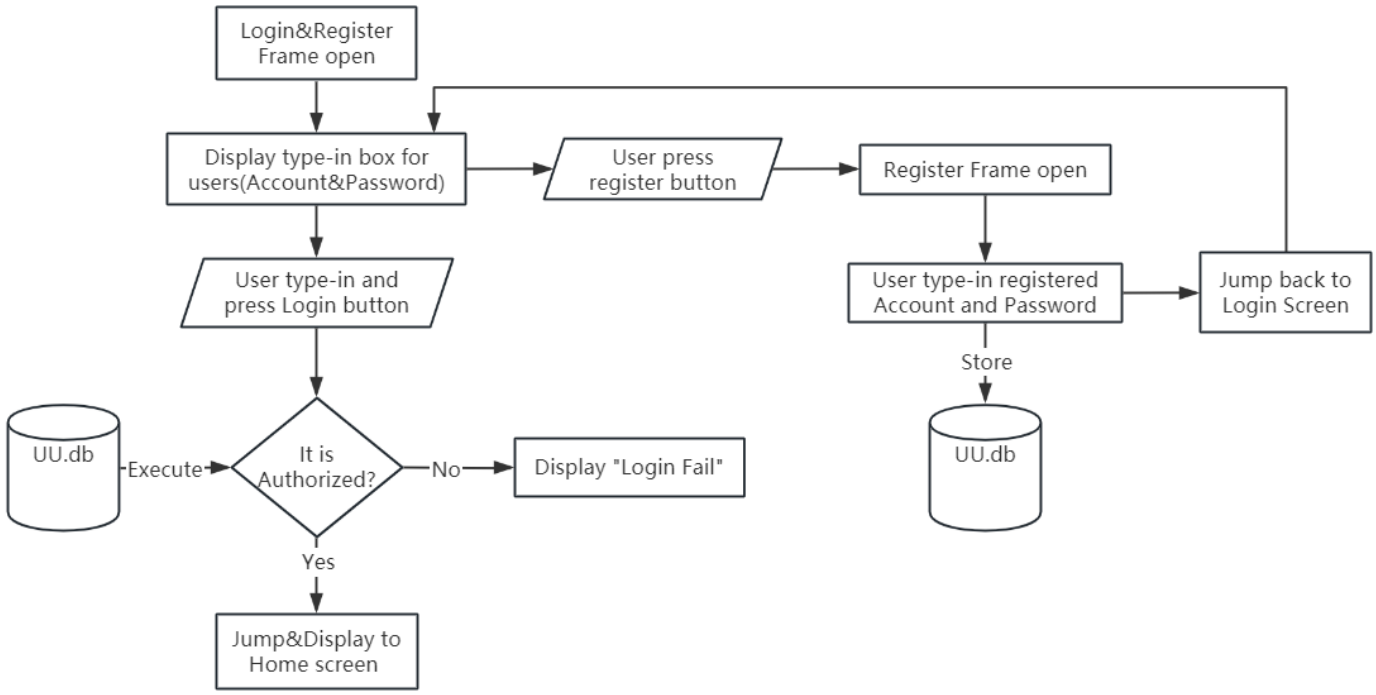
Web-Frame Flowcharts:

These flowcharts display the connection between different website and how they can be accessed.



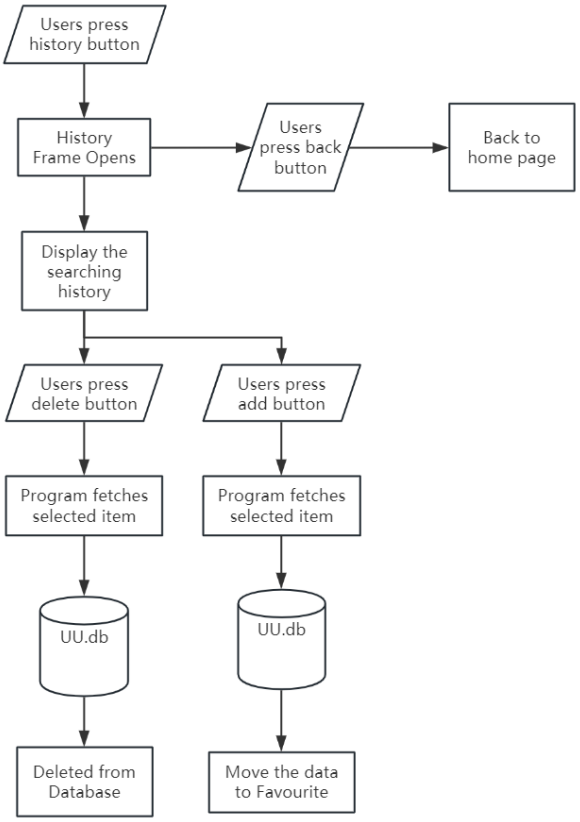
Login & Register:

The system will check whether user's account is correct from database, if it's not, the webpage will display alert.



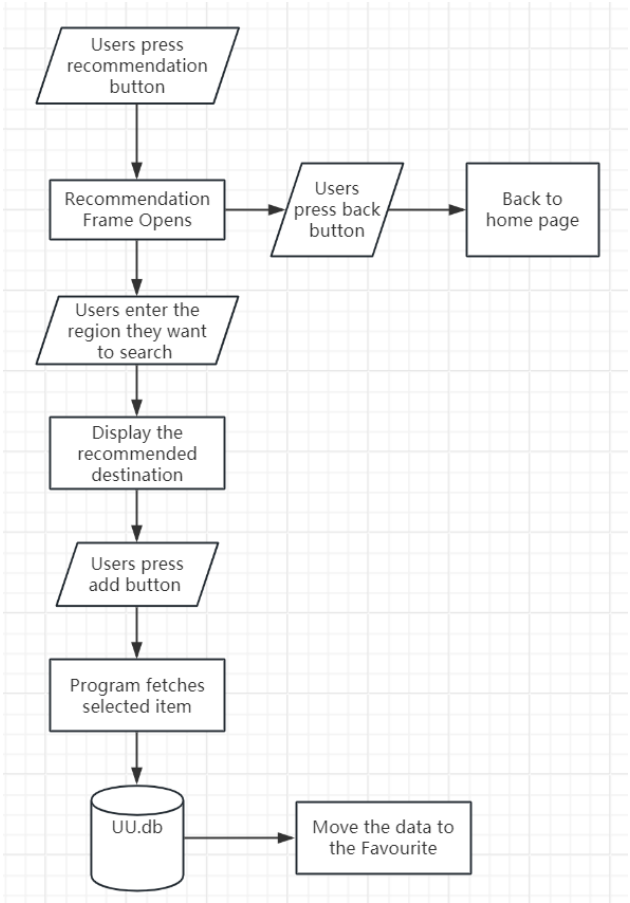
The following flowcharts represent the control of Web-Frame in more detail.

History:



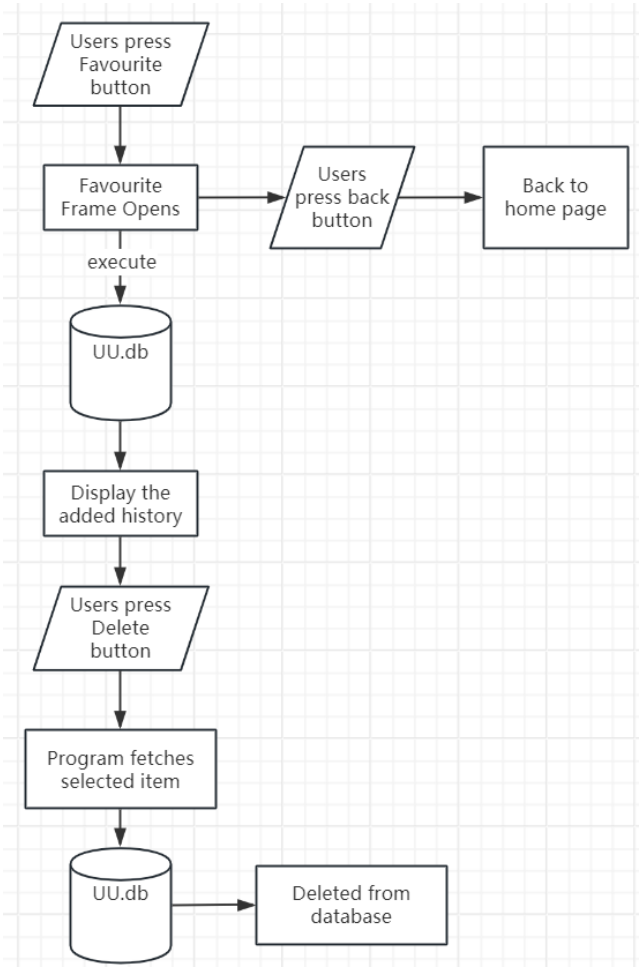
The webpage will send a route and require the history data store in database, and display in a form of table.

Recommendation:



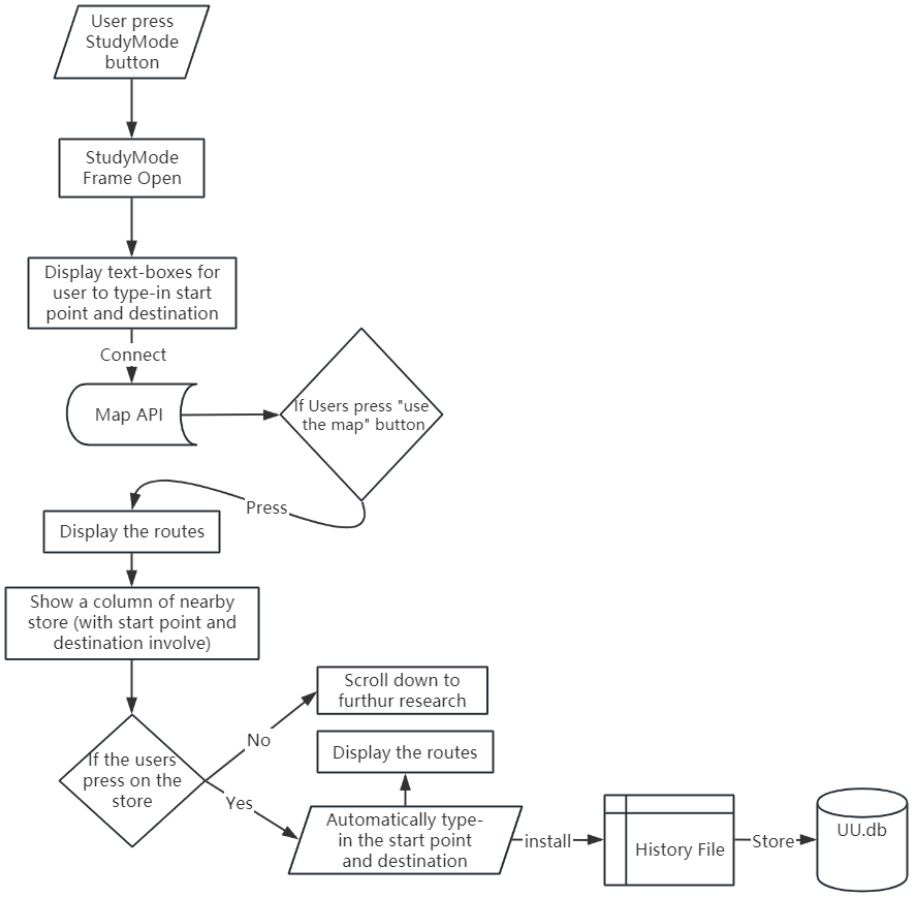
Users can search their liked destination, and store the destination into the database.

Favourite:



The stored liked destination by users will be display in a form of table through this webpage.

StudyMode:



The system will connect to map API, and implement some special functions to design the route for users.

MYSQL Database structures:

favourite	
	`idd` INT NOT NULL AUTO_INCREMENT FIRST, ADD PRIMARY KEY (`idd`);
	`tipinput` VARCHAR(255) NULL DEFAULT NULL
	`time` VARCHAR(255) NOT NULL DEFAULT CURRENT_TIMESTAMP

history	
	`idd` INT NOT NULL AUTO_INCREMENT FIRST, ADD PRIMARY KEY (`idd`);
	`tipinput` VARCHAR(255) NULL DEFAULT NULL
	`time` VARCHAR(255) NOT NULL DEFAULT CURRENT_TIMESTAMP

travellinghistory	
	`idd` INT NOT NULL AUTO_INCREMENT FIRST, ADD PRIMARY KEY (`idd`);
	`tipinput` VARCHAR(255) NULL DEFAULT NULL
	`tepinput` VARCHAR(255) NULL DEFAULT NULL
	`topinput` VARCHAR(255) NULL DEFAULT NULL
	`twpinput` VARCHAR(255) NULL DEFAULT NULL
	`choice` VARCHAR(255) NULL DEFAULT NULL
	`time` VARCHAR(255) NOT NULL DEFAULT CURRENT_TIMESTAMP

ap_list	
	`id` INT NOT NULL AUTO_INCREMENT FIRST, ADD PRIMARY KEY (`id`);
	`account` VARCHAR(255) NULL DEFAULT NULL
	`password` VARCHAR(255) NULL DEFAULT NULL

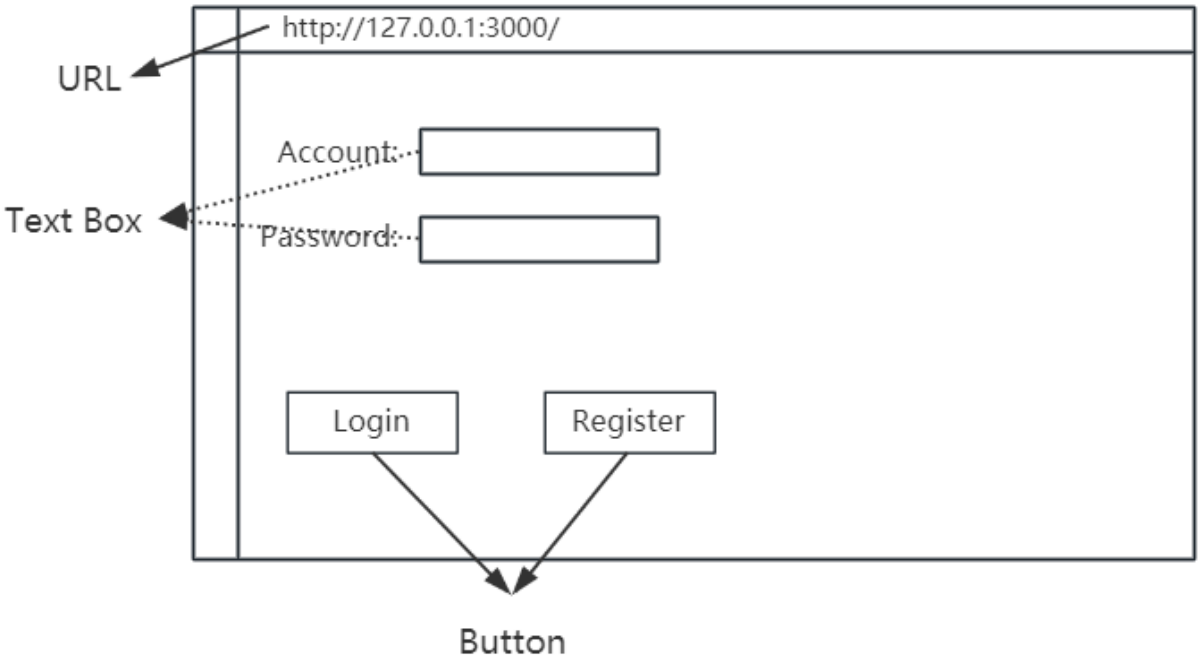
Under a database, these four tables will be created. For ap_list to other tables, they are one-to-many relation, as one user won't only have one piece of searched history or liked destination.

Program Design:

The following graphs are the parts of GUI that are going to be d splayed within the program to operate Webpage.

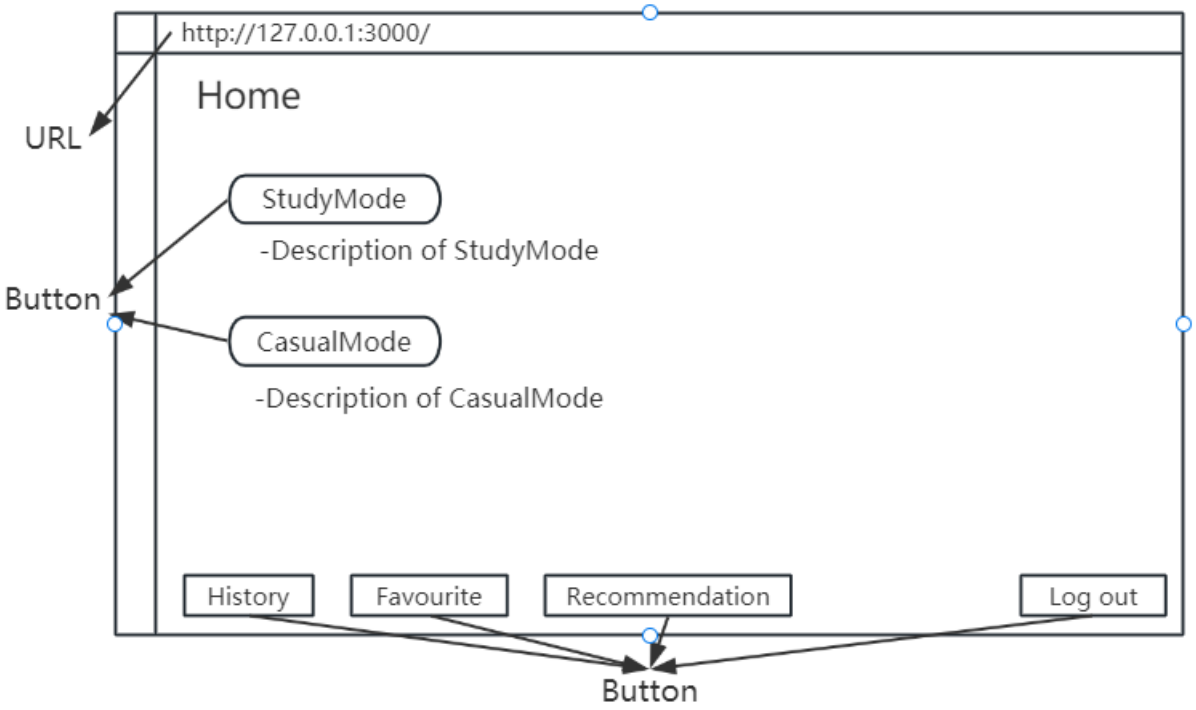
Login Screen Layout:

The text box is for users to type in their prefer accounts and passwords. If they don't have an exist account, they can choose to register a new account.



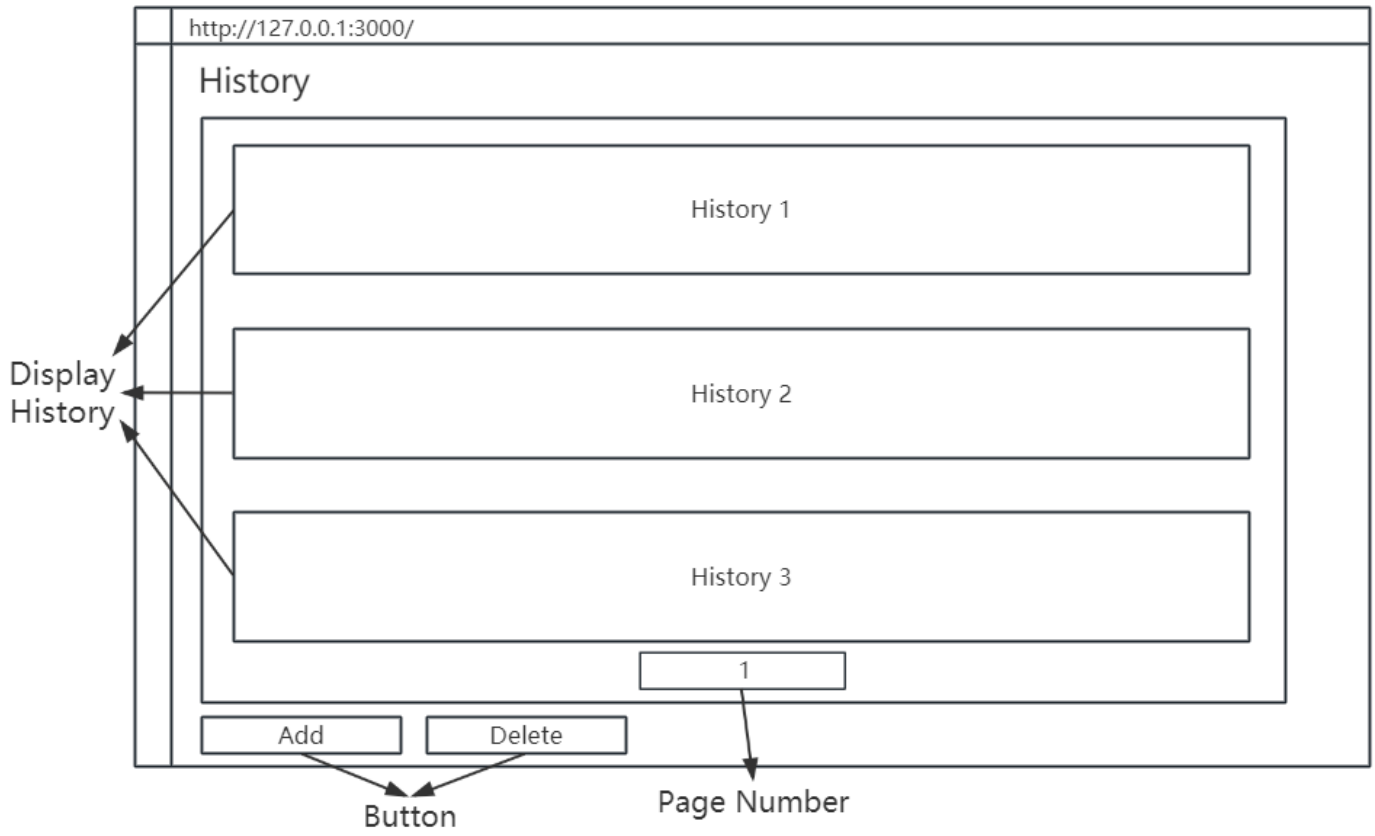
Home Screen Layout:

The two buttons are for users to choose their preferential mode aiming for different purpose during their routes. For instance, if the users choose StudyMode, the follow-up routes will search out library or academic related stores or shops, provide and show to the users. For CasualMode, the system will recommend some food stores or entertainment to the users' preferences. On beneath of the modes, those buttons are used to check the research history, the stores that they are preferred and some recommended destinations or stores. Or for users to log out the account.



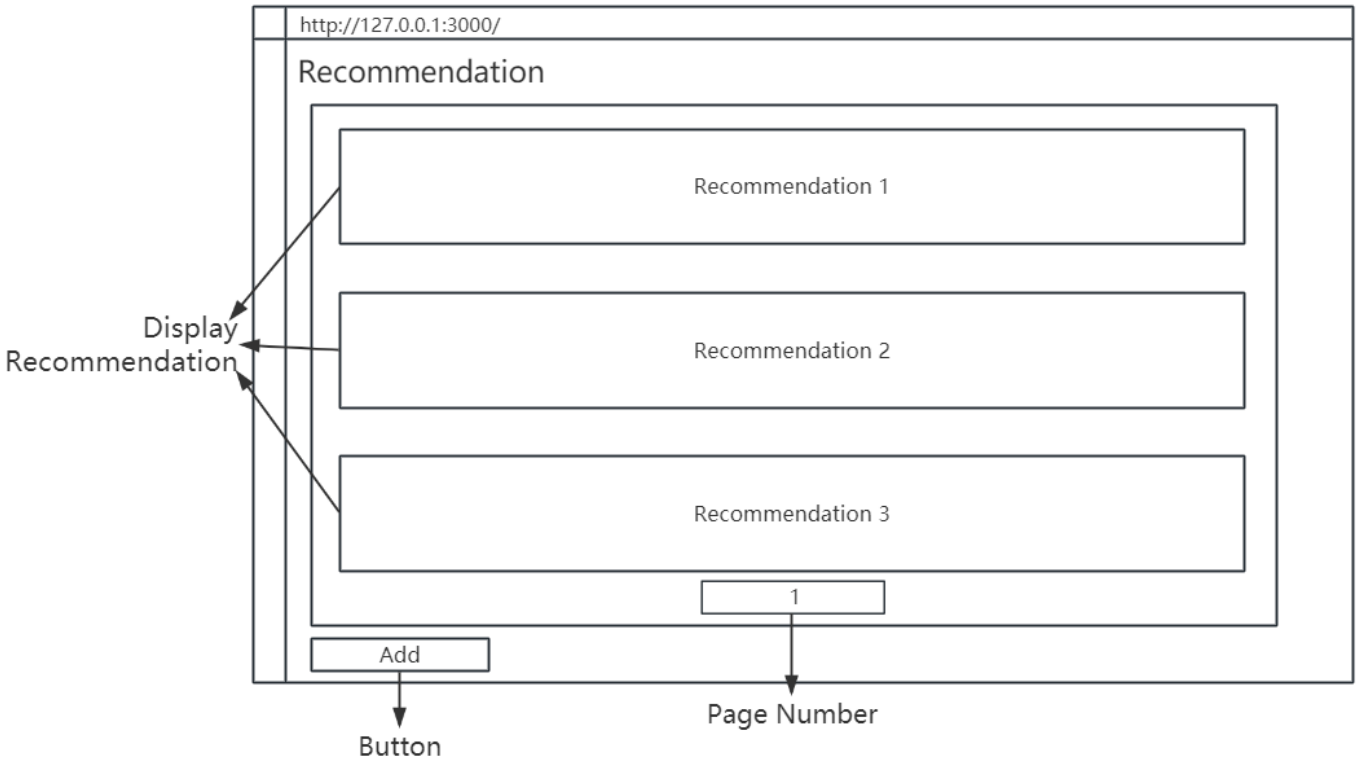
History Screen Layout:

This screen displays the history Webpage. The ‘Add’ button is used to add the history into the favorite database. For ‘Delete’ button is used to click the history and delete the history forever.



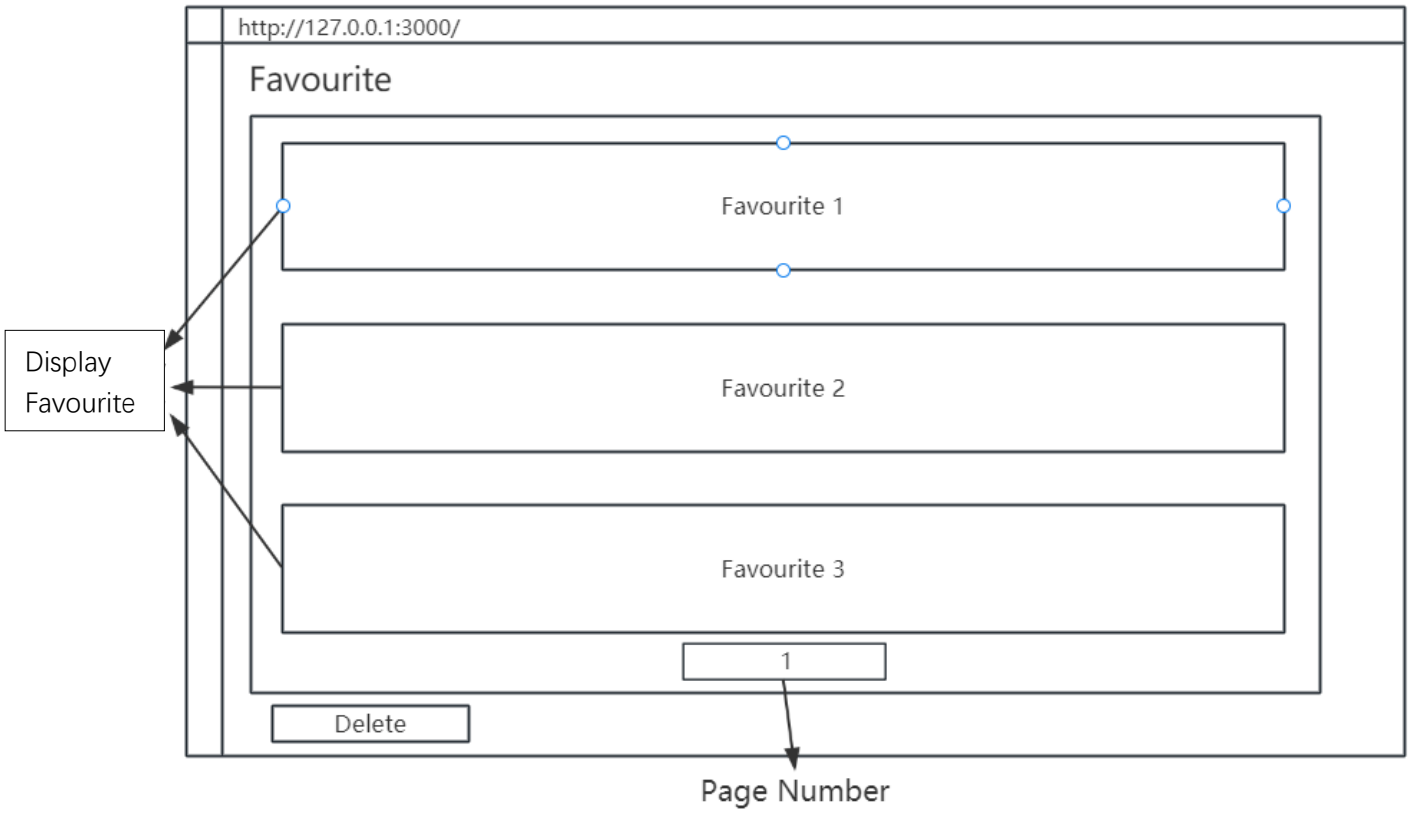
Recommendation Screen Layout:

This screen displays the recommendation Webpage. The ‘Add’ button is also used to add the recommendation stores into the favorite table in database.



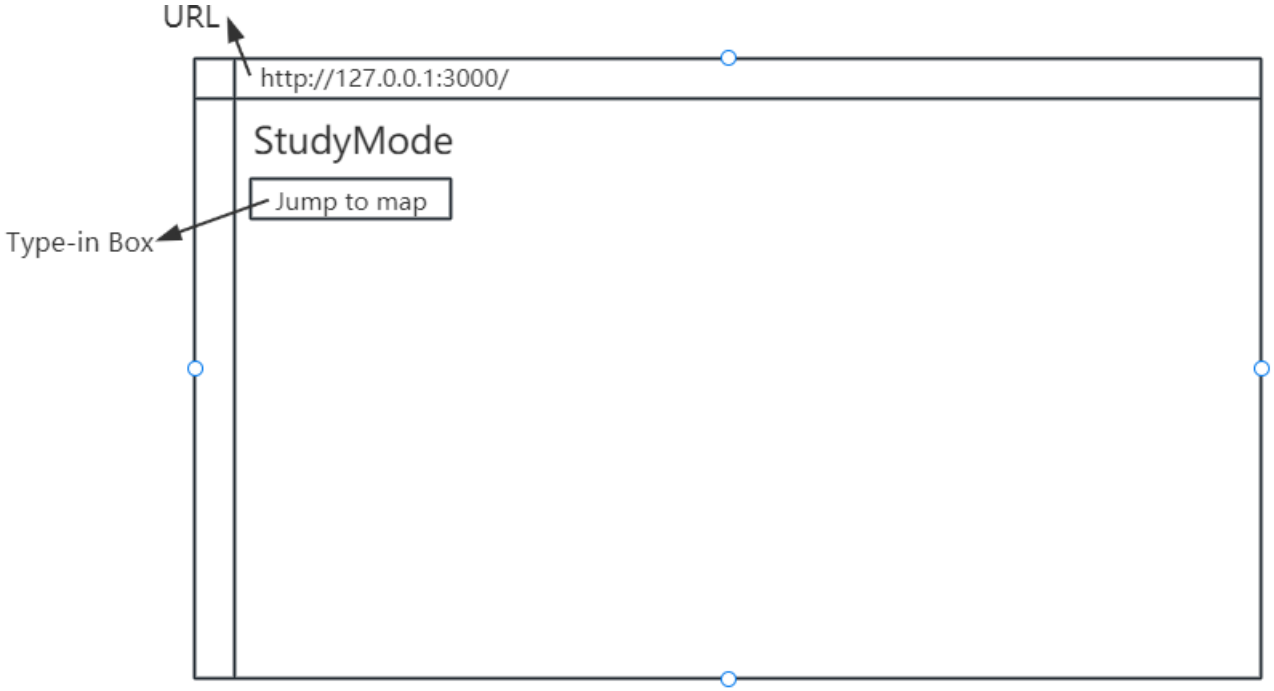
Favourite Screen Layout:

This screen indicates the Favourite Webpage. The ‘Delete’ button is used to delete forever from the favorite.



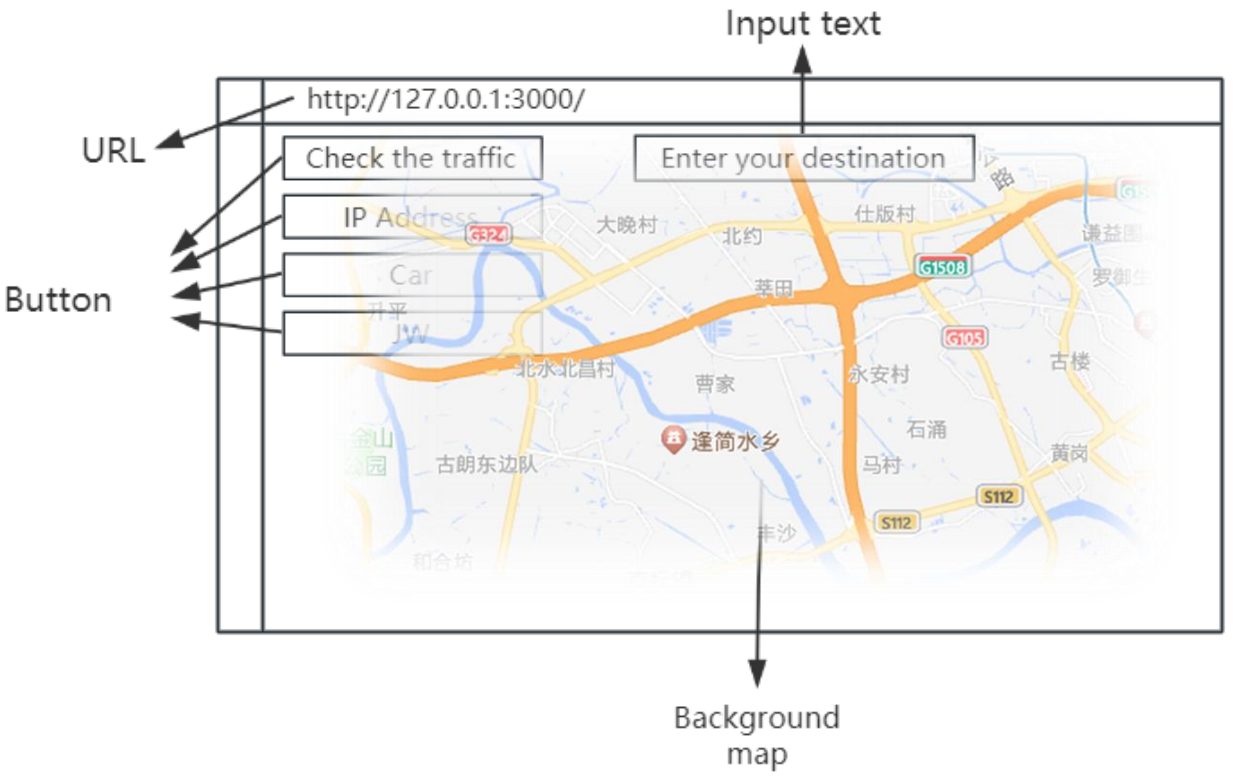
StudyMode Screen Layout:

This screen indicates the StudyMode Map Webpage. The ‘Jump to map’ button is used to link the webpage of the users and connect the code of StudyMode preferences. And use the map for navigation.



Map Home Screen Layout:

The screen indicates the Map Home Webpage. ‘Check the traffic’ button will help the users to jump to another map to check the current traffic. ‘IP address’ is used to check whether the IP address of the users is workable. ‘Car’ button is used to navigate the cars’ routes. ‘JW’ button is used to check the latitude and longitude of the users. The input text box is used to check detailed information for users to check anywhere in China they’d like.



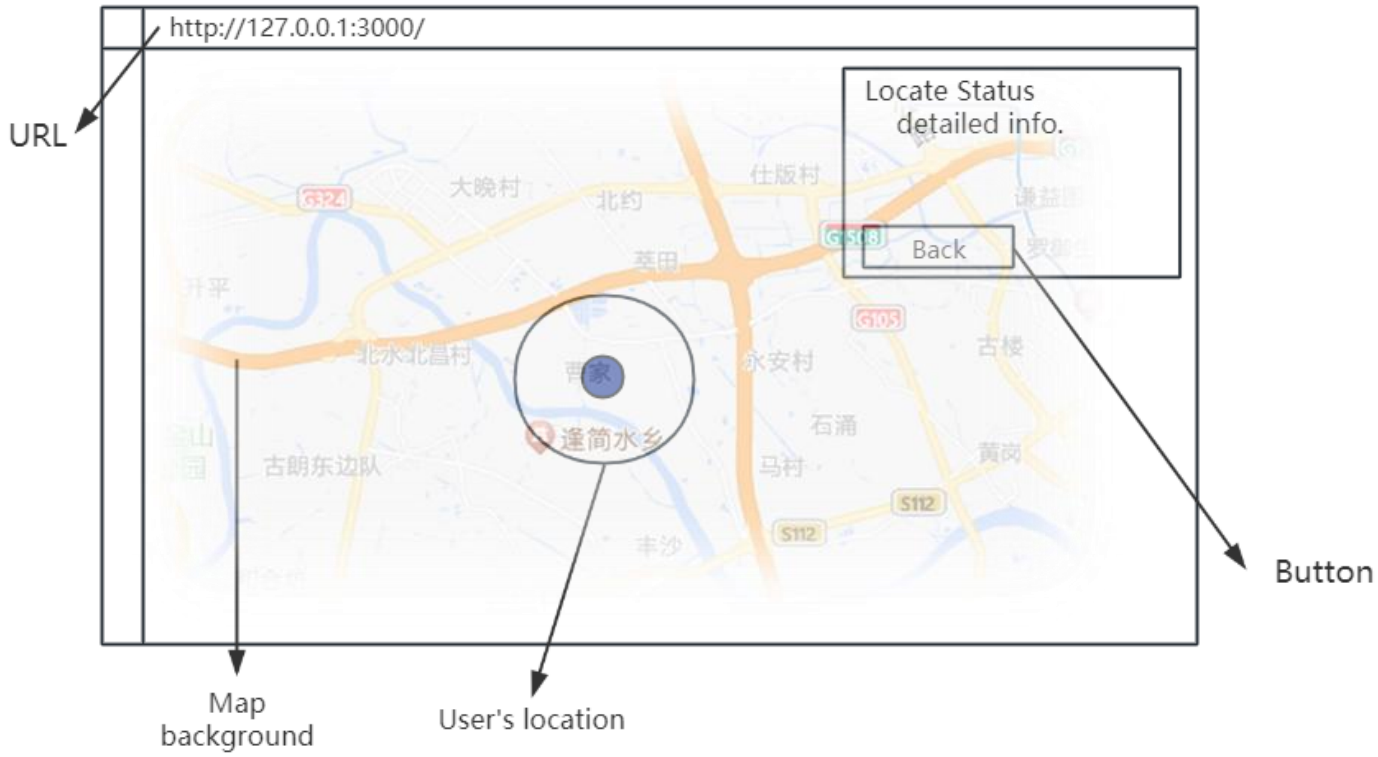
Check the traffic Screen Layout:

The screen indicates the traffic Webpage. ‘Hidden’ is used to hide the traffic lines, ‘Shows’ is to show. ‘Refresh’ is used to refresh the traffic map. ‘Back’ is used to turn back to the previous Webpage.



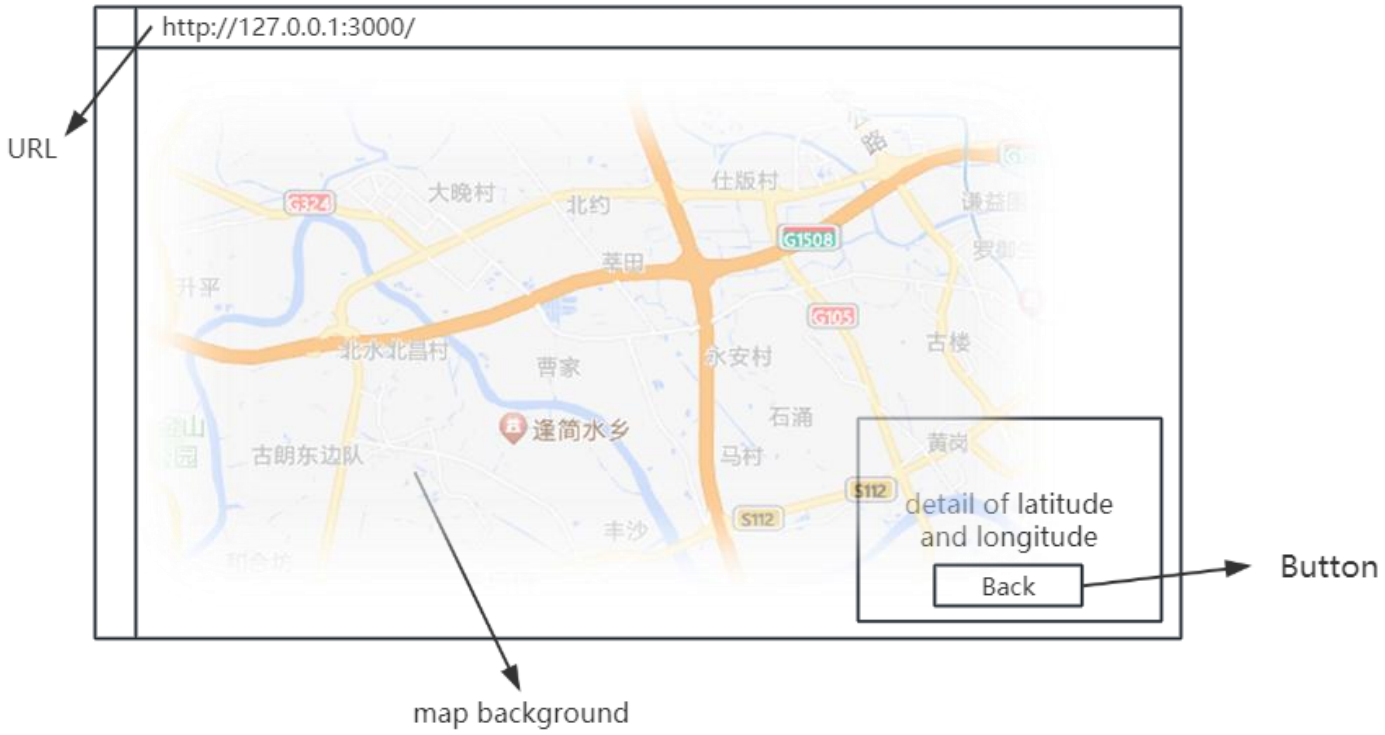
Check the IP Address of Users Screen Layout

The screen indicates the IP address of users Webpage. The circle presents the current location of users. Users can clearly know their position and what are some places they can enjoy nearby. The detailed information will also demonstrate the real-time weather in users' district.



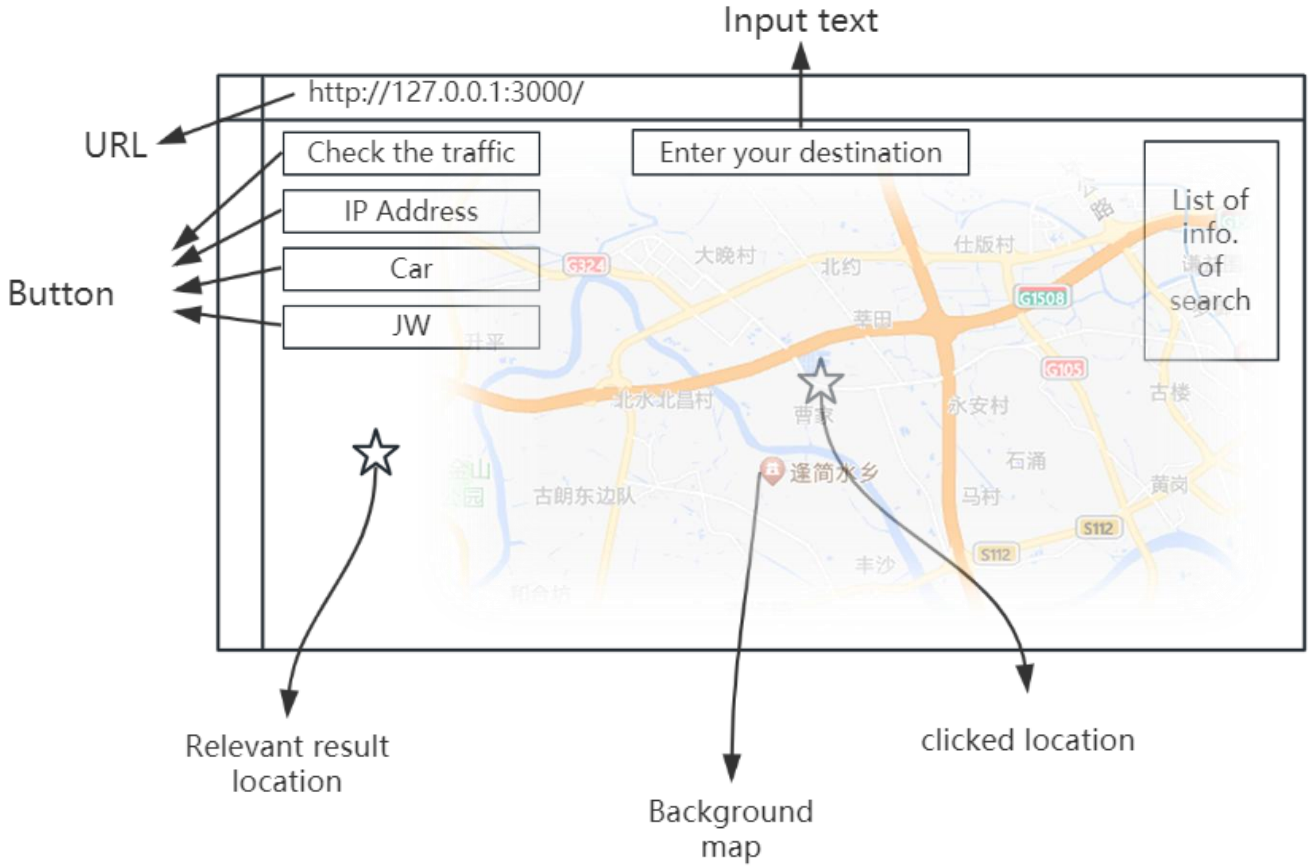
Record the latitude & longitude of start point and destination

The screen indicates the latitude and longitude of location Webpage.



The information of searching specific destination

The screen shows the information that the client search. After entering the destination, and press ‘enter’, the map will show a route and a list of details around the route.



Test Plan:

Test Type	Example
Working website	When the server.js is opened, the whole website should be run and can be get access through browsers.
Ensure the authentication is accurate	If a user has already entered their stored credentials in the database, they should be redirected to the home page. When users enter wrong credentials, the website should be able to give out warnings. If user does not have an account, they should have a choice to register new account.
Error Handling (data types)	All the users-input in the text filed should be entered the relative data types correctly.
Error Handling (MYSQL)	All the errors related to connection to the MYSQL database should be handled within the code, and output related warning.
Enable editing	All the data present in the history data should enable users to edit or delete it.
Data redundancy	Data should not be used or stored unnecessarily; all the data should prevent to have multiple copies of it.